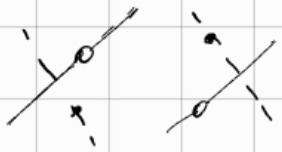
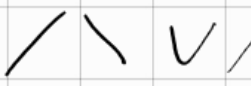
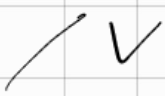
Т1 $x^2 = y^2$

а) Сколько ф-ий $y: R \to R$ удовл?  Более, чем счетное

б) Сколько неч ф-ий $y: R \to R$ удовл?  4

в) Сколько неч ф-ий $y: R \to R$ удовл и $y(1)=1$?  2

г) Сколько неч ф-ий $y: [1;2] \to R$ удовл и $y(1)=1$  1

§3 №61(2)

$x\cos y + y\cos u + u\cos x = 1 \quad (0;1;0)$

$F'_u = -y\sin u + \cos x = 1$

$F'_x = \cos y - u\sin x = \cos 1$

$F'_y = -x\sin y + \cos u = 1$

$u'_x = -\frac{F'_x}{F'_u} = -\cos 1$

$u'_y = -\frac{F'_y}{F'_u} = -1$

§3 №64(1)

$u^3 - xu + y = 0$

а) $(3;-2;2)$ $u'_x = -\frac{F'_x}{F'_u} = \frac{u}{3u^2-x} = \frac{2}{12-3} = \frac{2}{9}$

$u'_y = -\frac{F'_y}{F'_u} = \frac{-1}{3u^2-x} = -\frac{1}{9}$

$du = \frac{2}{9}dx - \frac{1}{9}dy$

б) $(3;-2;-1)$ $F'_u = 3u^2 - x = 3-3 = 0 \Rightarrow$ не существует du

§3 №72

$f(x-az; y-bz) = 0 \quad f(u;v)$ - дифф. $\quad a; b = const \quad z(x;y)$

$u = x - az$

$v = y - bz$

$f(u;v) = 0$

$f'_u(1 - az'_x) + f'_v(-bz'_x) = 0$

$f'_u(-az'_y) + f'_v(1 - az'_y) = 0$

$$f'_u = (af'_u + bf'_v)z'_x$$
$$f'_v = (af'_u + bf'_v)z'_y$$
$$af'_u + bf'_v = (af'_u + bf'_v)(az'_x + bz'_y)$$
$$az'_x + bz'_y = 1 \Rightarrow a\frac{\partial z}{\partial x} + b\frac{\partial z}{\partial y} = 1$$

§3 №75 (1;2) $xe^{u+v} + 2uv = 1$

$ye^{u-v} - \frac{u}{1+v} = 2x$

$u(1;2) = v(1;2) = 0$

$F_1(x;y;u;v) = xe^{u+v} + 2uv - 1 = 0$

$F_2(x;y;u;v) = ye^{u-v} - \frac{u}{1+v} - 2x = 0$

$$\frac{\partial \vec{F}}{\partial \vec{y}} = \begin{pmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} \end{pmatrix}\Bigg|_{\dots} = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix} \qquad \det = -3$$

$$\frac{\partial F_1}{\partial u} = xe^{u+v} + 2v = 1 \qquad \frac{\partial F_1}{\partial v} = xe^{u+v} + 2u = 1$$

$$\frac{\partial F_2}{\partial u} = ye^{u-v} - \frac{1}{1+v} = 1 \qquad \frac{\partial F_2}{\partial v} = -ye^{u-v} + \frac{u}{(1+v)^2} = -2$$

$$\left(\frac{\partial \vec{F}}{\partial \vec{y}}\right)^{-1} = \begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix} \cdot \frac{-1}{3} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$$

$$\left(\frac{\partial F}{\partial \vec{x}}\right)\Big|_{\dots} = \begin{pmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{pmatrix}\Bigg|_{\dots} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$$

$$\frac{\partial F_1}{\partial x} = e^{u+v} = 1 \qquad \frac{\partial F_1}{\partial y} = 0 \qquad \frac{\partial F_2}{\partial x} = -2 \qquad \frac{\partial F_2}{\partial y} = e^{u-v} = 1$$

$$\vec{f}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u(x;y) \\ v(x;y) \end{pmatrix}$$

$$\frac{\partial \vec{f}}{\partial \vec{x}} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = -\frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = -\frac{1}{3}\begin{pmatrix} 0 & 1 \\ 3 & -1 \end{pmatrix}$$

$$\frac{\partial u}{\partial x} = 0 \qquad \frac{\partial u}{\partial y} = -\frac{1}{3} \qquad \frac{\partial v}{\partial x} = 1 \qquad \frac{\partial v}{\partial y} = \frac{1}{3}$$

§4 №436(6) $u(1;1) = 0$

$F(x;y;u) = y - u - e^{xu} = 0$

$F'_x = -ue^{xu}\big|_{\dots} = 0$

$F'_y = 1$

$F'_u = -1 - xe^{xu}\big|_{\dots} = -2$

$du = \frac{1}{2}dy$

$u'_x = \frac{-ue^{xu}}{1+xe^{xu}}\Big|_{1;1;0} = 0$

$u'_y = \frac{1}{1+xe^{xu}}\Big|_{1;1;0} = \frac{1}{2}$

$F''_{xx} = -u'_x e^{xu} - uxe^{xu} = 0$

$F''_{ux} = -e^{xu} - xue^{xu}u'_x = -1$

$F''_{xy} = -u'_y e^{xu} - uxe^{xu}u'_y = -\frac{1}{2}$

$F''_{uy} = -x^2 e^{xu}u'_y = -\frac{1}{2}$

$F''_{yy} = 0$

$F''_{uy} = -x^2 e^{xu}u'_y = -\frac{1}{2}$

$$u''_{xx} = \left(-\frac{F'_x}{F'_u}\right)'_x = \frac{-F''_{xx}F'_u + F'_x F''_{ux}}{(F'_u)^2} = 0$$

$$u''_{xy} = \left(-\frac{F'_x}{F'_u}\right)'_y = \frac{-F''_{xy}F'_u + F'_x F''_{uy}}{(F'_u)^2} = \frac{-1+0}{4}$$

$$u''_{yy} = \left(-\frac{F'_y}{F'_u}\right)'_y = \frac{-F''_{yy}F'_u + F'_y F''_{uy}}{(F'_u)^2} = \frac{0-\frac{1}{2}}{4} = -\frac{1}{8}$$

$$d^2u = -\frac{1}{2}dx\,dy - \frac{1}{8}dy^2$$

§ 46 (2)

$$f\left(\frac{x}{u};\frac{y}{u}\right) = 0$$

$f(a;b)=0 \quad \frac{\partial f}{\partial a} = f'_1 \quad \frac{\partial f}{\partial b} = f'_2$

$$F'_x = f'_1 \cdot \frac{1}{u}$$

$$F'_y = f'_2 \cdot \frac{1}{u}$$

$$F'_u = f'_1\left(-\frac{x}{u^2}\right) + f'_2\left(-\frac{y}{u^2}\right)$$

$$u'_x = \frac{f'_1 u}{xf'_1 + yf'_2} \qquad u'_y = \frac{f'_2 u}{xf'_1 + yf'_2}$$

$$F''_{xx} = \frac{f''_{11}}{u^2} \qquad F''_{xy} = \frac{f''_{12}}{u^2} \qquad F''_{yy} = \frac{f''_{22}}{u^2}$$

$$F''_{xu} = \frac{-xf''_{11} - yf''_{12} - uf'_1}{u^3}$$

$$F''_{yu} = \frac{-xf''_{12} + yf''_{12} + uf'_2}{u^3}$$

$$u''_{xx} = \left(-\frac{F'_x}{F'_u}\right)'_x = \frac{-F''_{xx}F'_u + F'_x F''_{ux}}{(F'_u)^2}$$

$$u''_{xy} = \left(-\frac{F'_x}{F'_u}\right)'_y = \frac{-F''_{xy}F'_y + F'_x F''_{uy}}{(F'_u)^2}$$

НУ ПОДСТАВЛЯТЬ — ЖЕСТЬ

$$u''_{yy} = \left(-\frac{F'_y}{F'_u}\right)'_y = \frac{-F''_{yy}F'_u + F'_y F''_{uy}}{(F'_u)^2}$$

$$d^2u = u''_{xx}dx^2 + 2u''_{xy}dx\,dy + u''_{yy}dy^2$$

§ 3 № 106 (1)

$u = xyz$

$V = xy - xyz$

$\omega = y - xy$

$$\begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial \omega}{\partial x} & \frac{\partial \omega}{\partial y} & \frac{\partial \omega}{\partial z} \end{pmatrix} =$$

$$= \begin{pmatrix} yz & xz & xy \\ y-yz & x-xz & -xy \\ -y & 1-x & 0 \end{pmatrix} = 0 + xy(1-x)(y-yz) + x^2y^2z + xy^2(1-z) + xy^2z(1-x) =$$

$= xy(y - xy - yz + xyz) + x^2y^2z + x^2y^2 - x^2y^2z + xy^2z - x^2y^2z =$

$= xy^2 - x^2y^2 - x^2y^2z + x^2y^2z + x^2y^2z + x^2y^2 - x^2y^2z + xy^2z - x^2y^2z = xy^2$

Т3

$f: \mathbb{R}^2 \to \mathbb{R}^2$

$u = e^x \cos y$

$v = e^x \sin y$

$J = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \begin{pmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{pmatrix}$ $\det J = e^{2x}\cos^2 y + e^{2x}\sin^2 y = e^{2x}$

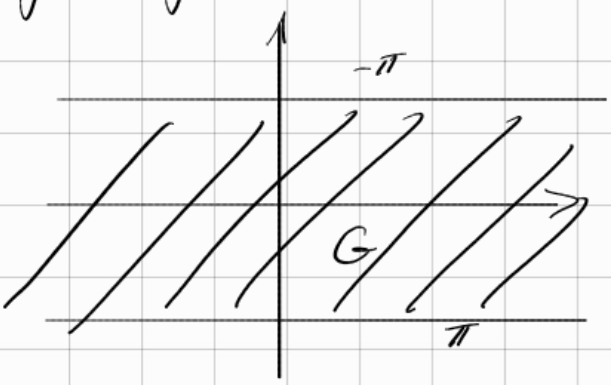
$\begin{pmatrix} u \\ v \end{pmatrix}(x;y) = \begin{pmatrix} u \\ v \end{pmatrix}(x; y+2\pi)$

не вз. однозначно на $\mathbb{R}^2$

Мн-во знач отображ $\mathbb{R}^2 \setminus (0;0)$

Образ $G = \mathbb{R}^2 \setminus (0;0)$

$\mathbb{R}^2 \setminus (0;0) \xleftrightarrow[\text{образ}]{\text{вз}} (-\infty; +\infty) \times [-\pi; \pi]$



Т4

$f: \mathbb{R}^2 \to \mathbb{R}^2$

$x = r\cos\varphi$

$y = r\sin\varphi$

$r > 0 \iff \varphi = \varphi(x;y)$, $r = r(x;y)$

$r = \sqrt{x^2+y^2}$

$\operatorname{tg}\varphi = \frac{y}{x}$

$\frac{\partial r}{\partial x} = \frac{2x}{2\sqrt{x^2+y^2}} = \frac{x}{r}$ $\quad \frac{\partial r}{\partial y} = \frac{y}{r}$

$\frac{1}{\cos^2\varphi}\varphi' = -\frac{y}{x^2}$ $\quad \varphi' = -\frac{y\cos^2\varphi}{x^2} = -\frac{y}{r^2}$

$\frac{1}{\cos^2\varphi}\varphi' = \frac{1}{x}$ $\quad \varphi' = \frac{\cos^2\varphi}{x} = \frac{x}{r^2}$

$J = \begin{pmatrix} r'_x & r'_y \\ \varphi'_x & \varphi'_y \end{pmatrix} = \begin{pmatrix} \frac{x}{r} & \frac{y}{r} \\ -\frac{y}{r^2} & \frac{x}{r^2} \end{pmatrix}$

§3 №6

$x\frac{\partial u}{\partial y} - y\frac{\partial u}{\partial x} = 0$

$x = r\cos\varphi$ $\quad \frac{\partial r}{\partial x} = \frac{x}{r}$ $\quad \frac{\partial r}{\partial y} = \frac{y}{r}$

$y = r\sin\varphi$ $\quad \frac{\partial \varphi}{\partial x} = -\frac{y}{r^2}$ $\quad \frac{\partial \varphi}{\partial y} = \frac{x}{r^2}$

$u = u(x;y) = \tilde{u}(r;\varphi) = \tilde{u}(r(x;y); \varphi(x;y))$

$\frac{\partial u}{\partial x} = \frac{\partial \tilde{u}}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial \tilde{u}}{\partial \varphi}\frac{\partial \varphi}{\partial x} = \frac{x}{r}\frac{\partial \tilde{u}}{\partial r} - \frac{y}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}$

$\frac{\partial u}{\partial y} = \frac{\partial \tilde{u}}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial \tilde{u}}{\partial \varphi}\frac{\partial \varphi}{\partial y} = \frac{y}{r}\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}$

Подставляем: $x\left(\frac{y}{r}\frac{\partial \tilde{u}}{\partial r}+\frac{x}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}\right)-y\left(\frac{x}{r}\frac{\partial \tilde{u}}{\partial r}-\frac{y}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}\right)=\frac{xy}{r}\frac{\partial \tilde{u}}{\partial r}+\frac{x^2}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}-\frac{xy}{r}\frac{\partial \tilde{u}}{\partial r}+\frac{y^2}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}=$

$=\frac{\partial \tilde{u}}{\partial \varphi}=0$

$\tilde{u}=f(r)\quad f\in C^1$ поч гр

$u=f(\sqrt{x^2+y^2})\quad f\in C$

§3 №88(2)

$x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=z \qquad x=u \quad y=uv$

$z(x;y)=z(u(x;y);v(x;y))$

$x=u \quad \frac{y}{x}=v$

$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial x}=\frac{\partial z}{\partial u}-\frac{\partial z}{\partial v}\frac{y}{x^2}$

$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial y}=\frac{\partial z}{\partial v}\frac{1}{x}$

$x\left(\frac{\partial z}{\partial u}-\frac{\partial z}{\partial v}\frac{y}{x^2}\right)+y\left(\frac{\partial z}{\partial v}\cdot\frac{1}{x}\right)=z$

$\frac{\partial z}{\partial u}u-\frac{\partial z}{\partial v}\cdot v+\frac{\partial z}{\partial v}\cdot v=z$

$\ln|z|=\ln|u|+C(v) \Rightarrow z=C(v)\cdot u$

$z=x\cdot C\left(\frac{y}{x}\right)$

§3 №90 $(z-x)\frac{\partial z}{\partial x}-y\frac{\partial z}{\partial y}=0$

$dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy$

$dx=\frac{\partial x}{\partial y}dy+\frac{\partial x}{\partial z}dz$

$dz=\frac{\partial z}{\partial x}\left(\frac{\partial x}{\partial y}dy+\frac{\partial x}{\partial z}dz\right)+\frac{\partial z}{\partial y}dy=\frac{\partial z}{\partial x}\frac{\partial x}{\partial z}dz+\left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial y}+\frac{\partial z}{\partial y}\right)dy$

$\frac{\partial z}{\partial x}\frac{\partial x}{\partial z}=1$

$\frac{\partial z}{\partial y}=-\frac{\partial z}{\partial x}\frac{\partial x}{\partial y}=-\frac{\frac{\partial x}{\partial y}}{\frac{\partial x}{\partial z}}$

$(z-x)\frac{1}{\frac{\partial x}{\partial z}}-y\left(-\frac{\frac{\partial x}{\partial y}}{\frac{\partial x}{\partial z}}\right)=0$

$z-x+y\frac{\partial x}{\partial y}=0$

$\frac{\partial x}{\partial y}=\frac{x-z}{y}$

§4 №51(1)

$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$

$x=r\cos\varphi$

$y=r\sin\varphi$

$u=u(x;y)=\tilde{u}(r;\varphi)=\tilde{u}(r(x;y);\varphi(x;y))$

$$\frac{\partial u}{\partial x} = \frac{\partial \tilde{u}}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial \tilde{u}}{\partial \varphi}\frac{\partial \varphi}{\partial x} = \frac{x}{r}\frac{\partial \tilde{u}}{\partial r} - \frac{y}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}$$

$$\frac{\partial u}{\partial y} = \frac{\partial \tilde{u}}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial \tilde{u}}{\partial \varphi}\frac{\partial \varphi}{\partial y} = \frac{y}{r}\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}$$

$$\frac{\partial^2 u}{\partial x^2} = \left(\frac{1}{r} - \frac{x^2}{r^3}\right)\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r}\left(\frac{\partial^2 \tilde{u}}{\partial r^2}\frac{x}{r} - \frac{\partial^2 \tilde{u}}{\partial r \partial \varphi}\frac{y}{r^2}\right) + \frac{2yx}{r^4}\frac{\partial \tilde{u}}{\partial \varphi} - \frac{y}{r^2}\left(\frac{\partial^2 \tilde{u}}{\partial r \partial \varphi}\frac{x}{r} - \frac{\partial^2 \tilde{u}}{\partial \varphi^2}\frac{y}{r^2}\right)$$

+

$$\frac{\partial^2 u}{\partial y^2} = \left(\frac{1}{r} - \frac{y^2}{r^3}\right)\frac{\partial \tilde{u}}{\partial r} + \frac{y}{r}\left(\frac{\partial^2 \tilde{u}}{\partial r^2}\frac{y}{r} + \frac{\partial^2 u}{\partial r \partial \varphi}\frac{x}{r^2}\right) - \frac{2yx}{r^4}\frac{\partial \tilde{u}}{\partial \varphi} + \frac{x}{r^2}\left(\frac{\partial^2 \tilde{u}}{\partial r \partial \varphi}\frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial \varphi^2}\frac{x}{r^2}\right)$$

$$\frac{1}{r}\frac{\partial \tilde{u}}{\partial r} + \frac{\partial^2 \tilde{u}}{\partial r^2} + \frac{\partial^2 \tilde{u}}{\partial \varphi^2}\frac{1}{r^2} = 0$$

§4 №52(1)

$$\frac{\partial^2 z}{\partial t^2} = a^2\frac{\partial^2 z}{\partial x^2}$$

$u = x - at$

$v = x + at$

$x = \frac{u+v}{2} \quad t = \frac{v-u}{2a}$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial t} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial t} = -a\frac{\partial z}{\partial u} + a\frac{\partial z}{\partial v}$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial u}\left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}\right)\frac{\partial u}{\partial x} + \frac{\partial}{\partial v}\left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}\right)\frac{\partial v}{\partial x} = \frac{\partial^2 z}{\partial u^2} + 2\frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}$$

$$\frac{\partial^2 z}{\partial t^2} = \frac{\partial}{\partial t}\left(a\left(\frac{\partial z}{\partial v} - \frac{\partial z}{\partial u}\right)\right) = a\frac{\partial}{\partial t}\left(\frac{\partial z}{\partial v} - \frac{\partial z}{\partial u}\right) = a\left(\frac{\partial}{\partial u}\left(\frac{\partial z}{\partial v} - \frac{\partial z}{\partial u}\right)\frac{\partial u}{\partial t} + \frac{\partial}{\partial v}\left(\frac{\partial z}{\partial v} - \frac{\partial z}{\partial u}\right)\frac{\partial v}{\partial t}\right) =$$

$$= a\left(\left(\frac{\partial^2 z}{\partial v \partial u} - \frac{\partial^2 z}{\partial u^2}\right)(-a) + \left(\frac{\partial^2 z}{\partial v^2} - \frac{\partial^2 z}{\partial u \partial v}\right)a\right) = a^2\left(\frac{\partial^2 z}{\partial u^2} - 2\frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}\right)$$

Подставим в исходное $\frac{\partial^2 z}{\partial u \partial v} = 0$

§5 №18(3)

$x^3 - y^2 + u^2 - 3x + uy + u - 8 = 0 \qquad u \neq -\frac{1}{2}$

$$u'_x = -\frac{F'_x}{F'_u} = -\frac{3x^2 - 3}{2u + 1}$$

$$u'_y = -\frac{F'_y}{F'_u} = -\frac{-2y + 4}{2u + 1}$$

$$\begin{cases} 3x^2 - 3 = 0 \\ -2y + 4 = 0 \end{cases} \begin{cases} x = \pm 1 \\ y = 2 \end{cases}$$

$(x;y) = (1;2) \quad u = \begin{bmatrix} -3 \\ 2 \end{bmatrix} \qquad (x;y) = (-1;2) \quad u = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

$$u''_{xx} = \frac{-6x(2u+1) + (3x^2 - 3)(2u'_x)}{(2u+1)^2} = 2$$

$$u''_{xy} = \frac{(3x^2 - 3)(2u'_y)}{(2u+1)^2} = 0$$

$$u''_{yy} = \frac{2(2u+1) + (-2y+u)(2u'_y)}{(2u+1)^2} = \frac{2}{3}$$

I) $(x;y;u) = (1;2;-3)$

$u''_{xx} = \frac{6}{5}$

$u''_{xy} = 0$

$u''_{yy} = -\frac{2}{5}$

II) $(x;y;u) = (1;2;2)$

$u''_{xx} = -\frac{6}{5}$

$u''_{xy} = 0$

$u''_{yy} = \frac{2}{5}$

III) $(x;y;u) = (-1;2;-2)$

$u''_{xx} = -2$ ЭКСТР. max

$u''_{xy} = 0$

$u''_{yy} = -\frac{2}{3}$

IV) $(x;y;u) = (-1;2;1)$

$u''_{xx} = 2$ ЭКСТР min

$u''_{xy} = 0$

$u''_{yy} = \frac{2}{3}$

§5 №21(2)

$u = 1 - 4x - 8y \quad x^2 - 8y^2 = 8$

$L = 1 - 4x - 8y + \lambda(x^2 - 8y^2 - 8)$

необх. усл.

$$\begin{cases} \frac{\partial L}{\partial x} = -4 + 2\lambda x = 0 \\ \frac{\partial L}{\partial y} = -8 - 16\lambda y = 0 \\ x^2 - 8y^2 - 8 = 0 \end{cases}$$

$\lambda \neq 0 \quad x = \frac{2}{\lambda} \quad y = -\frac{1}{2\lambda}$

$\frac{4}{\lambda^2} - 8\frac{1}{4\lambda^2} = 8$

$\frac{2}{\lambda^2} = 8 \quad \lambda = \pm\frac{1}{2}$

1) $\lambda = \frac{1}{2} \quad (4;-1)$

2) $\lambda = -\frac{1}{2} \quad (-4;1)$

Достат. усл. $\frac{\partial^2 L}{\partial x^2} = 2\lambda \quad \frac{\partial^2 L}{\partial x \partial y} = 0 \quad \frac{\partial^2 L}{\partial y^2} = -16\lambda$

$$d^2L = (dx\ dy) \begin{pmatrix} 2\lambda & 0 \\ 0 & -16\lambda \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix} = 2\lambda\, dx^2 - 16\lambda\, dy^2 = 2\lambda(dx^2 - 8dy^2)$$

$d\varphi = 0$

$\varphi = x^2 - 2y^2 - 8$

$2x\,dx - 4y\,dy = 0$

$dx = \frac{2y}{x}\,dy$

$\Rightarrow 2\lambda\left(64\frac{y^2}{x^2} - 8\right)dy^2$

$d^2L|_{\lambda} = 2\cdot\frac{1}{2}\left(64\cdot\frac{1}{16} - 8\right)dy = -4\,dy^2 < 0 \Rightarrow (4;-1)$ – т. max $u = -7$

$d^2L|_{\lambda} = 2\left(-\frac{1}{2}\right)\left(64\cdot\frac{1}{16} - 8\right)dy = 4dy^2 > 0 \Rightarrow (4;1)$ – т. min $u = 9$

§5 №25 (8)

$$\begin{cases} u = xy + 2xz + 2yz \\ xyz = 108 \end{cases}$$

$L = xy + 2xz + 2yz + \lambda(xyz - 108)$

$$\begin{cases} \frac{\partial L}{\partial x} = y + 2z + \lambda yz = 0 \\ \frac{\partial L}{\partial y} = x + 2z + \lambda xz = 0 \\ \frac{\partial L}{\partial z} = 2x + 2y + \lambda xy = 0 \\ xyz - 108 = 0 \end{cases} \Rightarrow x = y = 2z$$

$4z^3 = 108$ $(x;y;z) = (6;6;3)$

$\lambda = 6 + 6 + \lambda\,18 = 0$

$\lambda = -\frac{12}{18} = -\frac{2}{3}$

Доп. усл

$\frac{\partial^2 L}{\partial x^2} = 0$ $\frac{\partial^2 L}{\partial x \partial y} = 1 + \lambda z = -1$ $\frac{\partial^2 L}{\partial x \partial z} = 2 + \lambda y = -2$

$\frac{\partial^2 L}{\partial y^2} = 0$ $\frac{\partial^2 L}{\partial y \partial z} = 2 + \lambda x = -2$ $\frac{\partial^2 L}{\partial z^2} = 0$

$d^2L = (dx\ dy\ dz)\begin{pmatrix} 0 & -1 & -2 \\ -1 & 0 & -2 \\ -2 & -2 & 0 \end{pmatrix}\begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} = -2\,dx\,dy - 4\,dx\,dz - 4\,dy\,dz < 0 \Rightarrow$ т. max

$u = 108$

№26 (2;3)

2) $$\begin{cases} u = xyz \\ xy + yz + zx = 8 \\ x + y + z = 5 \end{cases}$$

$z = 5 - x - y$

$L = xyz + \lambda(xy + yz + zx - 8) + \mu(x + y + z - 5)$

$$\begin{cases} \frac{\partial L}{\partial x} = yz + \lambda(y+z) + \mu = 0 \\ \frac{\partial L}{\partial y} = xz + \lambda(x+z) + \mu = 0 \\ \frac{\partial L}{\partial z} = xy + \lambda(x+y) + \mu = 0 \\ xy + yz + xz = 8 \\ x + y + z = 5 \end{cases} \Rightarrow \begin{matrix} (y-x)(z+\lambda) = 0 \\ (z-y)(x+\lambda) = 0 \\ (x-z)(y+\lambda) = 0 \end{matrix}$$

в силу симметрии: $x = z$

$2x + y = 5 \quad y = 5 - 2x$

$3x^2 - 10x + 8 = 0 \quad x = \begin{bmatrix} 2 \\ \frac{4}{3} \end{bmatrix}$

1) $(x;y;z) = (2;1;2)$ (и все перестановки)

$u = 4$

2) $(x;y;z) = \left(\frac{4}{3}; \frac{7}{3}; \frac{4}{3}\right)$ (и все перестановки)

$u = \frac{112}{27}$

$t^3 - (x+y+z)t^2 + (xy+yz+zx)t - xyz = 0$

$t^3 - 5t^2 + 8t - u = 0$

$f(t) = t^3 - 5t^2 + 8t$

$f'(t) = 3t^2 - 10t + 8$

$t_1 = 2 \quad t_2 = \frac{4}{3}$

$f(2) = 4$ - лок. мин.

$f\left(\frac{4}{3}\right) = \frac{112}{27}$ - лок. макс.

чтобы 3 ве з корня $4 \le u \le \frac{112}{27}$

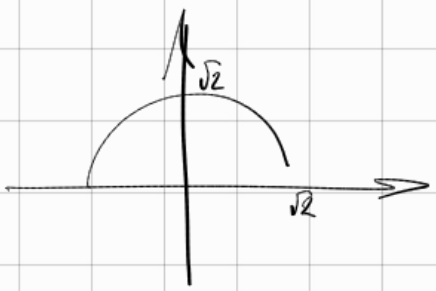
$\Downarrow$

min при $u = 4$ $(2;1;2)$ $(1;2;2)$ $(2;2;1)$

max при $u = \frac{112}{27}$ $\left(\frac{4}{3};\frac{7}{3};\frac{4}{3}\right)$ $\left(\frac{7}{3};\frac{4}{3};\frac{4}{3}\right)$ $\left(\frac{4}{3};\frac{4}{3};\frac{7}{3}\right)$

3) $\begin{cases} u = xy + yz \\ x^2 + y^2 = 2 \\ y + z = 2 \\ y \ge 0 \end{cases}$

$\begin{cases} z = 2 - y \\ u = xy + 2y - y^2 \\ x^2 + y^2 = 2 \\ y \ge 0 \end{cases}$



$x = \sqrt{2}\cos\varphi$

$y = \sqrt{2}\sin\varphi$

$\varphi \in (0;\pi)$

$u = 2\cos\varphi\sin\varphi + 2\sqrt{2}\sin\varphi - \sin^2\varphi$

$u'_\varphi = 2\cos 2\varphi + 2\sqrt{2}\cos\varphi - 4\sin\varphi\cos\varphi = 0$

$2\cos^2\varphi - 1 + \sqrt{2}\cos\varphi - 2\sin\varphi\cos\varphi = 0$

$$\sin\varphi = \frac{2\cos^2\varphi + \sqrt{2}\cos\varphi - 1}{2\cos\varphi}$$

$$\frac{4\cos^2\varphi + 2\cos^2\varphi + 1 - 4\cos^2\varphi - 2\sqrt{2}\cos\varphi + 4\sqrt{2}\cos^3\varphi}{4\cos^2\varphi} + \cos^2\varphi = 1$$

$$8\cos^4\varphi - 6\cos^2\varphi + 4\sqrt{2}\cos^3\varphi - 2\sqrt{2}\cos\varphi + 1 = 0$$

$$8t^4 + 4\sqrt{2}t^3 - 6t^2 - 2\sqrt{2}t + 1 = 0$$

$z = t\sqrt{2} \quad t = \frac{z}{\sqrt{2}}$

$$2z^4 + 2z^3 - 3z^2 - 2z + 1 = 0$$

$$(z^2-1)(2z^2+2z-1) = 0$$

$$\begin{cases} z = \pm 1 \\ z = \frac{-1 \pm \sqrt{3}}{2} \end{cases}$$

$$\begin{cases} t = \pm\frac{1}{\sqrt{2}} \\ t = \frac{-1 \pm \sqrt{3}}{2\sqrt{2}} \end{cases} \qquad \begin{cases} \cos\varphi = \pm\frac{1}{\sqrt{2}} \\ \cos\varphi = \frac{-1 \pm \sqrt{3}}{2\sqrt{2}} \end{cases} \qquad (x;y) = \begin{cases} (1;1) \\ (-1;1) \\ \left(\frac{-1-\sqrt{3}}{2}; \frac{\sqrt{3}-1}{2}\right) \\ \left(\frac{-1+\sqrt{3}}{2}; \frac{1+\sqrt{3}}{2}\right) \end{cases}$$

$$u''_{\varphi\varphi} = -4\sin\varphi\cos\varphi + 4\cos\varphi(-\sin\varphi) + 2\sqrt{2}(-\sin\varphi) - 4\cos\varphi\cos\varphi - 4\sin\varphi(-\sin\varphi) = -4\sin 2\varphi - 2\sqrt{2}\sin\varphi - 4\cos 2\varphi$$

$\varphi = \frac{\pi}{4}$

$u''_{\varphi\varphi} = -6 < 0 \Rightarrow ((1;1)$ т. max

$\varphi = \frac{3\pi}{4}$

$u''_{\varphi\varphi} = 2 > 0 \Rightarrow (-1;1;1)$ т. min

§5 №31 (3)

$u = x + y + z \qquad x^2 + y^2 \le z \le 1$

1) Проверим экстремумы внутри:

$\nabla u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \Rightarrow$ нет экстр внутри области

1. "Крышка" $z = 1 \quad x^2 + y^2 \le 1$

$u = x + y + 1 \qquad x^2 + y^2 \le 1$

внутри круга $\nabla u = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

на границе $u = x + y + 1 \quad x^2 + y^2 = 1$

$x = \cos\varphi \quad y = \sin\varphi$

$u = \cos\varphi + \sin\varphi + 1$

$u' = -\sin\varphi + \cos\varphi = 0$

$\sin\varphi = \cos\varphi \qquad \varphi = \frac{\pi}{4} + \pi k$

$(x;y) = \left(\frac{1}{\sqrt{2}}; \frac{1}{\sqrt{2}}\right) \quad u = 1 + \sqrt{2} \qquad x;y = \left(\frac{1}{\sqrt{2}}; -\frac{1}{\sqrt{2}}\right) \quad u = 1 - \sqrt{2}$

2. "Таблетка"

$x^2+y^2=z$ $\quad u=x+y+z$

$x^2+y^2 \le 1$

$x^2+y^2=z$ $\quad x^2+y^2<1$

$x^2+y^2=z$ $\quad x^2+y^2=1$ граница круга

$u = x+y+x^2+y^2$

$x^2+y^2<1$

$\nabla u = \begin{pmatrix} 1+2x \\ 1+2y \end{pmatrix} = 0$ $\quad \begin{cases} x=-\frac{1}{2} \\ y=-\frac{1}{2} \end{cases}$

$\Lambda = 1+\sqrt{2}$

$m = -\frac{1}{2}$

$u = -\frac{1}{2} - \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = -\frac{1}{2}$